

Mini SIEM Threat Detection Platform – Software Requirements Specification

Software Requirements Specification (SRS)
Mini SIEM Threat Detection Platform
Version: 1.0
Date: March 2026

1. INTRODUCTION

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the Mini SIEM Threat Detection Platform. The document provides guidance for developers, security engineers, DevOps engineers, QA teams, SOC analysts, and project stakeholders.

The platform aims to provide centralized log management, threat detection, security monitoring, and incident response capabilities comparable to enterprise SIEM tools.

1.2 Scope

The Mini SIEM platform will provide:

- Centralized log ingestion
- Log normalization and enrichment
- Real-time and batch threat detection
- Alert generation
- Case management
- Threat hunting capabilities
- Dashboards and reporting
- Role-based administration

The platform will not:

- Replace endpoint protection tools
- Replace firewall or IDS/IPS
- Perform packet-level network inspection
- Act as a vulnerability scanner

1.3 Definitions, Acronyms, and Abbreviations

SIEM – Security Information and Event Management

SOC – Security Operations Center

EPS – Events Per Second

RBAC – Role Based Access Control

IOC – Indicator of Compromise

CEF – Common Event Format

TLS – Transport Layer Security

API – Application Programming Interface

MFA – Multi-Factor Authentication

GeoIP – Mapping IP address to geographic location

DLQ – Dead Letter Queue

SLA – Service Level Agreement

1.4 References

- IEEE 830 Software Requirements Specification Standard
- NIST Cybersecurity Framework
- MITRE ATT&CK Framework
- OWASP Security Guidelines
- ISO 27001 Security Controls

1.5 Document Overview

Section 1 – Introduction

Section 2 – Overall System Description

Section 3 – Functional Requirements

Section 4 – External Interfaces

Section 5 – Non-Functional Requirements

Section 6 – Traceability Matrix

Section 7 – Appendices

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Mini SIEM platform acts as a centralized security analytics system collecting logs from infrastructure, applications, and network devices.

Security Architecture Flow:

Log Sources → Log Collectors → Message Queue → Detection Engine → Alerts → Dashboard

2.2 User Characteristics

SOC Analyst (Level 1)

Responsibilities:

- Monitor alerts
- Investigate suspicious activity
- Escalate incidents

Security Engineer

Responsibilities:

- Create detection rules
- Tune detection logic
- Maintain log pipelines

Auditor / CISO

Responsibilities:

- Review compliance reports
- Assess security posture
- Analyze incident trends

System Administrator

Responsibilities:

- Deploy agents
- Maintain infrastructure
- Ensure system uptime

2.3 Operating Environment

Deployment environments include:

- Cloud infrastructure
- On-premise data centers
- Hybrid deployments

Infrastructure technologies may include containerization and distributed search and streaming systems.

2.4 Design and Implementation Constraints

- Backend services implemented in Python or Go
- Web interface using modern JavaScript frameworks
- Distributed event streaming architecture
- Horizontal scalability required
- Minimum throughput 5,000 EPS

2.5 Assumptions and Dependencies

- Log sources produce structured logs
- Reliable network connectivity exists
- Message broker and storage clusters are available

3. SPECIFIC FUNCTIONAL REQUIREMENTS

3.1 Log Collection Module

- FR■LC■1: System shall support agent-based log collection.
- FR■LC■2: System shall support agentless syslog ingestion.
- FR■LC■3: System shall support Syslog, HTTP, TCP, and UDP protocols.
- FR■LC■4: System shall support JSON, CLF, and Windows Event Logs.
- FR■LC■5: Log ingestion must use TLS encryption.
- FR■LC■6: Authentication required using API keys or mutual TLS.
- FR■LC■7: System shall process at least 5,000 events per second.
- FR■LC■8: Logs shall be queued if downstream systems are unavailable.

3.2 Log Parsing & Normalization Module

- FR■LN■1: System shall normalize logs into a unified schema.
- FR■LN■2: Mandatory normalized fields: timestamp, source_ip, destination_ip, event_type, severity, hostname, log_source.
- FR■LN■3: Vendor-specific logs must map to common schema.
- FR■LN■4: Parsing failures shall be sent to a Dead Letter Queue.
- FR■LN■5: Parsing errors shall be logged.

3.3 Detection Engine Module

- FR■DE■1: Users shall create detection rules via UI.
- FR■DE■2: Rule types include threshold, correlation, absence, behavioral.
- FR■DE■3: System shall support real-time detection.
- FR■DE■4: System shall support batch analysis.
- FR■DE■5: Rules may map to MITRE ATT&CK tactics.
- FR■DE■6: Rule testing environment shall exist.

3.4 Alerting & Case Management Module

FR■AC■1: Alert severity levels: Critical, High, Medium, Low, Info.
FR■AC■2: Alerts enriched with GeoIP and threat intelligence.
FR■AC■3: Notifications supported: dashboard, email, webhook.
FR■AC■4: Alerts automatically create investigation cases.
FR■AC■5: Case lifecycle states: Open, Investigating, Contained, Resolved, Closed.
FR■AC■6: Analysts may mark alerts as false positives.

3.5 Dashboard & Visualization Module

FR■DV■1: Provide real■time security dashboard.
FR■DV■2: Display alert counts by severity.
FR■DV■3: Provide attack timeline visualization.
FR■DV■4: Show top attacking IPs.
FR■DV■5: Provide geolocation attack maps.
FR■DV■6: Display system health metrics.

3.6 Threat Hunting Module

FR■TH■1: Analysts can search logs.
FR■TH■2: Queries support filters (IP, user, event type, timeframe).
FR■TH■3: Query results should return within 2 seconds.
FR■TH■4: Users may save searches.
FR■TH■5: Search results exportable.

3.7 Reporting Module

FR■RP■1: Scheduled reports supported.
FR■RP■2: Compliance report templates provided.
FR■RP■3: Export formats include PDF, CSV, JSON.
FR■RP■4: Custom report builder available.

3.8 Administration Module

FR■AD■1: System supports user management.
FR■AD■2: Role■based access control enforced.
FR■AD■3: Administrators manage log sources.
FR■AD■4: Administrators configure detection policies.
FR■AD■5: All administrative actions logged.

4. EXTERNAL INTERFACE REQUIREMENTS

User Interface

- Web dashboard
- Responsive design
- Browser support (Chrome, Firefox, Edge)

Hardware Interface

- Linux servers
- Windows servers
- Container environments

Software Interface

- Search engine integration
- Streaming platform integration
- Threat intelligence feeds

Communication Interface

- REST APIs
- WebSockets for real■time updates
- HTTPS secure transport

5. NON■FUNCTIONAL REQUIREMENTS

Performance

- Minimum ingestion: 5,000 EPS
- Query response <2 seconds (95th percentile)
- Dashboard load <3 seconds
- Alert generation latency <5 seconds

Scalability

- Horizontal scaling supported
- Auto■scaling triggers based on load
- Storage retention up to 365 days

Security

- Single Sign■On support
- Multi■Factor Authentication
- TLS encryption
- Role■based access control
- Audit logging

Reliability

- System uptime: 99.5%
- Daily backups
- Disaster recovery capability

Maintainability

- Centralized logging
- Configuration management
- Automated upgrades

Usability

- User documentation
- Analyst training materials
- Accessibility standards compliance

6. REQUIREMENTS TRACEABILITY MATRIX

FR■LC■1 → Business Objective: Centralized Logging → Test Case: TC■01

FR■DE■1 → Business Objective: Threat Detection → Test Case: TC■12

FR■AC■1 → Business Objective: Incident Response → Test Case: TC■21

FR■TH■1 → Business Objective: Threat Hunting → Test Case: TC■31

7. APPENDICES

Use Case Examples

Use Case: Investigate Alert

Actor: SOC Analyst

Steps:

1. Analyst logs into dashboard
2. System displays alerts
3. Analyst reviews alert
4. Analyst opens investigation case
5. Analyst escalates if necessary

Use Case: Create Detection Rule

Actor: Security Engineer

Steps:

1. Engineer creates rule
2. Defines conditions
3. Maps to ATT&CK tactic
4. Saves rule
5. System activates detection

Data Flow Overview

Log Sources → Log Collectors → Message Queue → Log Parser
 → Detection Engine → Alert Manager → Dashboard / Reports